Checkpoint 3 — Identify the Posterior for a Conjugate Pair — Student Version

INFO 521 — Machine Learning Foundations · checkpoint (prompts & criteria; graded in class)

> ℹ How this works
>
> This checkpoint is sat **in class, closed-book** (~15–20 minutes, one derivation) and graded Satisfactory / Not-Yet against the criteria below. Version B is the alternate-version retake, offered the following week; you keep your best result. Knowing the prompt in advance is the point — practice until you can reproduce the derivation unaided.

Sitting: Week 4 · retake: Week 5 · gates Project 1.3.

Gates: Project 1.3 (Bayesian) **Format:** in-class, closed-book, ~15 min. Graded Satisfactory / Not-Yet. **Retakeable.**

This checkpoint tests the study-guide outcome *"given a conjugate prior–likelihood pair, identify the form of the posterior."* It uses Beta–Binomial as the clean, time-boxed instance. The project applies the **same principle** to the Gaussian–Gaussian pair — make that connection explicit when you hand it back.

Prompt (Version A)

A coin has unknown heads probability r. You place a prior $r \sim \mathrm{Beta}\,(\alpha, \beta)$ and observe y heads in N tosses (Binomial likelihood). Identify the posterior distribution of r — name the family and give its parameters. State in one sentence what makes this pair *conjugate*.

Prompt (Version B — retake)

You place a Gaussian prior $w \sim \mathcal{N}(0, \tau^2)$ on a single weight in a Gaussian-noise linear model. Without full derivation, name the family of the posterior $p(w \mid \mathrm{data})$ and explain why conjugacy guarantees it stays in that family.

Satisfactory criteria

- Names the correct posterior family.
- (A) Gives correct updated parameters.
- States conjugacy = prior and posterior share a family (posterior stays analytically tractable).